
Cheat Sheet — Project 13: Text to Speech (one-page carry card)

Course: 192.151 Introduction to Deep Learning · 2026S · Project 13 · **Option C (VITS)** · TU Wien **Team:** Dima (model & training) · Emir (infra, data, engineering) · Ilya (evaluation) **Repo:** github.com/Dmitri1S9/DL · 🗿 dataset `Dmi1tr13/ljspeech-b1` · checkpoints `Dmi1tr13/vits-b1-droid`

The one-liner

We fine-tuned a pretrained VITS model onto a **custom B1 droid voice** we built with RVC. The timbre moved to the droid (MCD ↓), at a measured cost to intelligibility (WER ↑).

The numbers

- **Base model:** `kakao-enterprise/vits-ljs` (single speaker, 22050 Hz)
- **Dataset:** LJSpeech (13,100 clips) → RVC → B1 droid → `Dmi1tr13/ljspeech-b1`. Held-out = **last 5%**.
- **Trained:** local **RTX 5060 (8 GB)**, **5 epochs**, **≈ 9.7 GPU-h** (+~3× in failed runs)
- **Config:** `lr 1e-4` · batch **1 + grad-accum** · AMP fp16 · segment 8192 · `grad-clip 10` · `disc-warmup 1500`
- **5 losses:** recon (L1 mel, **x45**) · KL · duration · adversarial (LSGAN) · feature-matching
- **Discriminator (from scratch):** MPD prime periods `(2, 7, 13, 29, 37, 73, 97, 113, 137)` · MSD 3 scales

Results (20 held-out clips, lower = better)

model	WER	CER	MCD (dB)	F0 RMSE (Hz)
base (pretrained)	9.1%	1.8%	10.48	30.8

model	WER	CER	MCD (dB)	F0 RMSE (Hz)
fine-tuned (droid)	29.2%	13.9%	6.19	32.4

Read it: MCD **10.5** → **6.2 dB** = learned the droid timbre (fine-tuning worked).
 WER **9** → **29%** = traded some clarity. Classic timbre-vs-intelligibility trade-off.

The debugging story (the best part)

Nothing converged at first. Three bugs: 1. **Duration loss ~100× too large** (averaged wrong) → `.sum()/mask.sum()` → dur **300** → **1.7** 2. **Grad-clip too tight** killed the generator update → loosen it → recon broke its plateau 3. **Fresh discriminator corrupted the decoder** → **warm up D 1500 steps** + lower LR → recon **0.85** → **0.40**, correct rhythm. (Lesson: a falling loss ≠ good speech; only inference WER caught it.)

Failure modes (honest)

- Intelligibility cost (WER up); some **short prompts collapse** (durations too short, e.g. “Roger roger...” → 0.8 s)
- Timbre ≠ hard robot: RVC moves **timbre**, not the ring-mod buzz (separate DSP effect)
- GAN artifacts early; over-fits past the best epoch
- Russian accent = **future work** (input pre-processing)
- **Future fix for WER:** freeze the text/timing front-end (keeps durations stable)

Demo (slide 16) — same sentence, before/after

“The mission is complete. All systems are nominal. Returning to base.” 1. **Base VITS** (pretrained) → 2. **Fine-tuned droid** (same line). Slide 17 = the collapse clip.

Q&A routing

Model/training → **Dima** · Evaluation → **Ilya** · Data/engineering → **Emir**.
 Repeat the question first.

Three things to remember

- MCD down = it learned the voice; WER up = the cost. That’s the headline.

- We **re-implemented VITS training from scratch** (HF ships inference-only): forward pass, MAS, 5 losses, discriminator, GAN loop.
- A converging loss does **not** prove good speech — that's why we evaluate the output (WER/CER/MCD/F0).